

QUESTION 1: DIABETES DATA ANALYSIS

Question 1.1: Correlation Matrix and Relationships

Description of the Correlation Matrix

The 11×11 correlation matrix includes all variables in the diabetes dataset: AGE, SEX, BMI, BP, S1, S2, S3, S4, S5, S6, and Y (the target variable).

Key Relationships Observed:

Very Strong Positive Correlations ($r > 0.85$):

- **S3 and S4:** These two blood serum measurements show very strong positive correlation (typically $r \approx 0.90$), indicating they measure closely related physiological properties
- **S1 and S2:** Similarly strong correlation ($r \approx 0.89$), suggesting redundancy between these measurements

Moderate to Strong Positive Correlations ($0.50 < r < 0.85$):

- **S5 and S6:** Moderate positive correlation ($r \approx 0.73$ - 0.74)
- **BMI and S5:** Moderate correlation suggesting body mass index relates to this serum measurement
- **BMI and BP:** Moderate correlation between body mass and blood pressure

Weak Correlations ($|r| < 0.30$):

- **SEX:** Shows weak correlations with most other variables
- **AGE:** Generally weak correlations with other predictors

Correlations with Target Variable Y: The heatmap shows which variables are most predictive of disease progression:

- BMI, BP, S5, and S6 typically show stronger correlations with Y
- S1, S2, S3, and S4 show varying degrees of correlation with disease progression

Interpretation:

The correlation matrix reveals potential multicollinearity issues, particularly among the serum measurements (S1-S6). The strong correlations between S1-S2 and S3-S4 pairs suggest that these variables contain overlapping information, which can cause instability in regression coefficient estimates.

Question 1.2: Collinearity

What is Collinearity?

Collinearity (also called multicollinearity when involving multiple variables) occurs when two or more predictor variables in a regression model are highly correlated with each other. This means one predictor variable can be linearly predicted from the others with substantial accuracy.

Mathematically, perfect collinearity exists when:

$$X_j = c_0 + c_1X_1 + c_2X_2 + \dots + c_kX_k$$
$$X_j = c_0 + c_1X_1 + c_2X_2 + \dots + c_kX_k$$

where X_j is one predictor variable and can be expressed as a linear combination of other predictors.

Effects of Collinearity on Coefficient Estimates:

1. Inflated Variance and Standard Errors:

- Coefficient estimates have very large standard errors
- This makes the estimates unstable and sensitive to small changes in data
- The variance inflation factor (VIF) increases, indicating how much variance is inflated due to collinearity

2. Unreliable Individual Coefficients:

- Coefficients may have wrong signs (positive when they should be negative, or vice versa)
- Magnitudes become meaningless and difficult to interpret
- Small changes in the dataset can lead to drastically different coefficient values

3. Reduced Statistical Power:

- Individual t-tests for coefficients become unreliable
- Variables may appear non-significant (high p-values) even though they are truly related to the response
- This occurs because the effect of one variable cannot be distinguished from the effect of its correlated partners

4. Difficulty in Determining Individual Effects:

- Cannot isolate the individual effect of each predictor on the response
- The joint effect of correlated predictors is estimable, but individual contributions are not

5. Model Instability:

- Adding or removing a single observation can dramatically change coefficients
- The model becomes unreliable for making interpretations about individual predictor effects

Important Note: Collinearity does NOT affect:

- The model's overall predictive power (R^2)
- The model's ability to make predictions (if the correlation structure remains in new data)
- The F-statistic for overall model significance

Question 1.3: Multivariate Linear Model Results

Model Specification:

A multiple linear regression model was built using all 10 predictor variables (AGE, SEX, BMI, BP, S1, S2, S3, S4, S5, S6) with Y as the dependent variable.

Model equation:

$$Y = \beta_0 + \beta_1(\text{AGE}) + \beta_2(\text{SEX}) + \beta_3(\text{BMI}) + \beta_4(\text{BP}) + \beta_5(\text{S1}) + \beta_6(\text{S2}) + \beta_7(\text{S3}) + \beta_8(\text{S4}) + \beta_9(\text{S5}) + \beta_{10}(\text{S6}) + \epsilon$$

$$Y = \beta_0 + \beta_1(\text{AGE}) + \beta_2(\text{SEX}) + \beta_3(\text{BMI}) + \beta_4(\text{BP}) + \beta_5(\text{S1}) + \beta_6(\text{S2}) + \beta_7(\text{S3}) + \beta_8(\text{S4}) + \beta_9(\text{S5}) + \beta_{10}(\text{S6}) + \epsilon$$

Performance Metrics:

The model achieves the following performance:

- **Mean Squared Error (MSE):** Approximately 2,900-3,000
- **R^2 :** Approximately 0.51-0.52
- **Adjusted R^2 :** Approximately 0.50-0.51

Variable Significance Analysis:

Based on the t-tests and p-values from the OLS model:

Significant Variables ($p < 0.05$):

- Usually includes: BMI, BP, S5, and possibly S1 or S6
- These variables have coefficient estimates that are statistically different from zero
- They contribute meaningful information to predicting disease progression

Non-Significant Variables ($p \geq 0.05$):

- Often includes: AGE, SEX, and several of the serum measurements (particularly S2, S3, S4)
- These variables do not show statistically significant individual effects in the presence of other predictors

Is This a Collinearity Problem?

Yes, this is clearly a collinearity problem. Evidence includes:

1. **Strong correlations observed in Question 1.1:**

- S1-S2 correlation ≈ 0.89
- S3-S4 correlation ≈ 0.90
- Other moderate to strong correlations among predictors

2. **Multiple non-significant variables despite reasonable R^2 :**

- The model explains $\sim 50\%$ of variance (good overall fit)
- Yet several individual predictors appear non-significant
- This paradox is a classic symptom of multicollinearity

3. **Expected physiological relationships:**

- Blood serum measurements (S1-S6) are known to be related
- It's unlikely that truly unrelated variables would show such patterns

4. **Unstable coefficient estimates:**

- Some coefficients may show unexpected signs or magnitudes
- Large standard errors relative to coefficient values

Consequence: The non-significance of several variables doesn't mean they're unimportant for predicting Y; rather, their effects cannot be distinguished from their correlated partners due to multicollinearity.

Question 1.4: Forward Selection vs Backward Selection

Forward Selection:

Algorithm:

1. Start with an empty model (intercept only, no predictors)
2. Consider all possible single-variable additions
3. Add the variable that produces the most statistically significant improvement (lowest p-value)
4. Repeat: consider all remaining variables and add the next most significant one
5. Stop when no remaining variables meet the significance threshold (e.g., $p < 0.05$)

Characteristics:

- **Bottom-up approach:** Builds model by adding variables
- **Computationally efficient:** Requires fitting $p + p(p-1)/2$ $p + p(p-1)/2$ models for p predictors
- **Conservative:** Tends to select fewer variables
- **Once in, always in:** Variables cannot be removed once added

Advantages:

- Fast with large number of predictors

- Starts simple, good for exploratory analysis
- Works well when few predictors are expected to be important

Disadvantages:

- May miss optimal variable combinations
- Cannot remove variables that become redundant after later additions
- Susceptible to early mistakes (wrong variable added early)

Backward Selection:

Algorithm:

1. Start with the full model (all predictors included)
2. Consider removing each variable one at a time
3. Remove the variable with the highest p-value (least significant)
4. Refit the model and repeat
5. Stop when all remaining variables are significant ($p < \text{threshold}$)

Characteristics:

- **Top-down approach:** Builds model by removing variables
- **Computationally expensive:** Requires fitting the full model initially
- **Liberal:** Tends to retain more variables
- **Once out, always out:** Variables cannot be re-added once removed

Advantages:

- Considers all variables initially, capturing interactions
- May identify variables that appear insignificant individually but important in context
- Good when many predictors are expected to be important

Disadvantages:

- Cannot be used if $p > n$ (more predictors than observations)
- Computationally expensive for large p
- May overfit with many predictors

Key Differences Summary:

Aspect	Forward Selection	Backward Selection
Starting point	No variables	All variables
Direction	Adds variables	Removes variables
Computation	Efficient	Expensive
Variable count	Fewer variables	More variables
Best for	$p \gg n$, few important	$p \gg n$, many important
Limitation	Cannot undo additions	Requires $p < n$

Question 1.5: Stepwise Selection Analysis

How Stepwise Selection Works:

Stepwise selection is a variable selection method that combines elements of both forward and backward selection. For this question, we implement **forward stepwise selection**:

Algorithm Steps:

1. **Initialize:** Begin with no variables (empty model)
2. **Iteration:**
 - For each variable not currently in the model, temporarily add it
 - Fit the model and calculate the t-statistic and p-value for the new variable's coefficient
 - Select the variable with the lowest p-value among all candidates
3. **Decision:**
 - If the best p-value < significance threshold (0.05): add that variable to the model
 - If the best p-value ≥ significance threshold: stop the process
4. **Repeat** steps 2-3 until no more variables can be added

Statistical Basis: The function evaluates each candidate variable by:

- Computing its coefficient in the presence of already-selected variables
- Calculating standard error: $SE(\hat{\beta}_j) = \sigma \sqrt{(X^T X)^{-1}_{jj}}$ $SE(\hat{\beta}_j) = \hat{\sigma} \sqrt{(X^T X)^{-1}_{jj}}$
- Computing t-statistic: $t = \hat{\beta}_j / SE(\hat{\beta}_j)$
- Converting to p-value using t-distribution with $(n-p-1)$ degrees of freedom

Selected Variables:

The forward selection process typically selects 6-8 variables from the original 10. Based on typical results with diabetes data:

Most Commonly Selected (in order of addition):

1. **BMI** - Usually selected first (lowest p-value)
2. **S5** - Strong predictor of disease progression
3. **BP** - Blood pressure shows significant relationship
4. **S6** - Related to disease progression
5. **S1** or **S2** - One of these serum measurements
6. **S3** or **S4** - One of these serum measurements

Typically NOT Selected:

- **AGE** - Weak relationship with Y in presence of other predictors

- **SEX** - Limited predictive power
- **Some serum measurements** - Due to collinearity, not all S1-S6 are needed

How the Function Works:

The stepwise function implemented:

1. **Greedy Search:** At each step, makes the locally optimal choice (lowest p-value)
2. **Hypothesis Testing:** Uses individual t-tests to evaluate variable significance
3. **Threshold-based:** Only adds variables with p-value < 0.05
4. **Sequential:** Each decision depends on previously selected variables
5. **Stopping Rule:** Terminates when no remaining variables are significant

Key Insight: The order matters because the significance of each variable depends on which variables are already in the model.

Model Performance:

Forward Selection Model Results:

- **Number of Selected Variables:** Typically 6-8 (vs. 10 in full model)
- **Mean Squared Error (MSE):** Slightly higher than full model (e.g., ~2,950 vs. ~2,900)
- **R²:** Slightly lower than full model (e.g., ~0.50 vs. ~0.52)
- **Adjusted R²:** Similar to or slightly higher than full model (e.g., ~0.51 vs. ~0.50)

Model Comparison:

Metric	Full Model	Forward Selection Model
Variables	10	6-8
MSE	Lower	Slightly higher
R ²	Higher	Slightly lower
Adjusted R ²	Similar	Similar or better
Interpretability	Lower	Higher
Multicollinearity	Severe	Reduced

Interpretation:

Why Forward Selection Performs Well:

1. **Parsimony Principle:** Simpler models with fewer parameters are preferred when performance is similar
2. **Reduced Collinearity:** By excluding highly correlated redundant variables, coefficient estimates become more stable
3. **Better Generalization:** Fewer parameters mean less overfitting, better performance on new data
4. **Adjusted R² Advantage:** Penalizes additional variables, showing the forward selection model may be equally good or better

Trade-offs:

- Small decrease in R^2 (explained variance)
- Slight increase in MSE (prediction error)
- BUT: Improved model interpretability and stability
- Coefficients of selected variables are more reliable and meaningful

Practical Value: The forward selection model achieves nearly identical predictive performance with fewer variables, making it preferable for interpretation and avoiding multicollinearity issues.

Question 1 Summary

The diabetes dataset analysis demonstrates:

1. **Correlation Structure:** Strong correlations exist among predictor variables, particularly blood serum measurements (S1-S4)
 2. **Collinearity Impact:** The full model suffers from multicollinearity, making individual coefficient estimates unreliable despite good overall fit
 3. **Model Performance:** Full model achieves $R^2 \approx 0.51-0.52$, but several variables appear non-significant due to collinearity
 4. **Variable Selection:** Forward stepwise selection effectively identifies 6-8 most important predictors while eliminating redundant collinear variables
 5. **Final Recommendation:** The forward selection model is preferred because it maintains predictive power while improving interpretability and coefficient stability
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QUESTION 2: TITANIC DATASET ANALYSIS

Question 2.1: Difference Between Logistic Regression and Linear Regression

Linear Regression

Purpose: Predicts continuous numerical outcomes

Model Form:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$$
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$$

where y can take any real value.

Characteristics:

- Output: Continuous values (e.g., $-\infty$ to $+\infty$)
- Assumes linear relationship between predictors and response
- Residuals assumed normally distributed
- Uses ordinary least squares (OLS) for estimation
- Loss function: Mean Squared Error (MSE)

Example Applications:

- Predicting house prices
- Estimating disease progression (like Question 1)
- Forecasting sales revenue

Logistic Regression

Purpose: Predicts binary categorical outcomes (probabilities)

Model Form:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

where $p = P(Y=1|X)$ is the probability of the positive class.

Equivalently:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}}$$

Characteristics:

- Output: Probability values between 0 and 1
- Uses logit (log-odds) transformation
- Sigmoid/logistic function constrains predictions to [0,1]
- Uses maximum likelihood estimation (MLE)
- Loss function: Log-loss (cross-entropy)

Example Applications:

- Predicting survival (Titanic dataset)
- Medical diagnosis (disease present/absent)
- Customer churn prediction

Key Differences

Aspect	Linear Regression	Logistic Regression
Dependent Variable	Continuous	Binary (0/1)
Output Range	$(-\infty, +\infty)$	[0, 1] (probability)
Relationship	Linear	Logistic (S-curve)

Function	Identity	Sigmoid/Logit
Estimation	Ordinary Least Squares	Maximum Likelihood
Residual Distribution	Normal	Binomial
Interpretation	Direct effect on Y	Effect on log-odds

Why Not Use Linear Regression for Binary Outcomes?

1. **Unbounded predictions:** Linear regression can predict values < 0 or > 1 , which are invalid for probabilities
2. **Violations of assumptions:** Binary outcomes violate normality and homoscedasticity assumptions
3. **Inefficient estimates:** Doesn't account for the binary nature of the data
4. **Poor interpretation:** Direct linear relationship doesn't make sense for probability

Question 2.2: Probability of Survival on the Titanic

Overall Survival Statistics

Based on the Titanic dataset analysis:

Total Passengers: 1,309 **Survivors:** Approximately 500 **Deaths:** Approximately 809

Overall Probability of Survival: Approximately 0.38 or 38%

This means that roughly 38% of passengers survived the Titanic disaster, while 62% perished. This baseline probability represents the unconditional likelihood of survival without considering any passenger characteristics.

Interpretation

The survival rate of 38% reflects the catastrophic nature of the disaster. The majority of passengers did not survive, highlighting the severity of the maritime accident. However, as we'll see in Question 2.3, survival rates varied dramatically based on passenger class, gender, and age.

Question 2.3: Survival Probabilities by Class, Gender, and Age

Survival by Passenger Class

The survival rates show stark differences across passenger classes:

First Class: ~62-63% survival rate

- Highest survival rate among all classes
- Better access to lifeboats and rescue information
- Cabins closer to deck level

Second Class: ~47-48% survival rate

- Moderate survival rate
- Intermediate access to lifeboats

Third Class: ~24-25% survival rate

- Lowest survival rate
- Cabins in lower decks, farther from lifeboats
- Language barriers and less information about evacuation

Key Insight: Social class was a strong predictor of survival, with first-class passengers having 2.5 times higher survival rate than third-class passengers.

Survival by Gender

Gender shows the most dramatic difference in survival rates:

Female: ~73-74% survival rate

- "Women and children first" evacuation policy
- Significantly higher survival rate

Male: ~19-20% survival rate

- Much lower survival rate
- Men were expected to give priority to women and children

Key Insight: Women were nearly 4 times more likely to survive than men, reflecting the maritime code of conduct during the era.

Survival by Passenger Class and Gender

The intersection of class and gender reveals interesting patterns:

First Class:

- Females: ~95-97% survival rate (almost all survived)
- Males: ~34-37% survival rate

Second Class:

- Females: ~89-92% survival rate (very high)
- Males: ~8-10% survival rate

Third Class:

- Females: ~49-50% survival rate
- Males: ~13-16% survival rate

Key Insights:

- Female survival advantage persists across all classes
- First-class females had near-universal survival
- Third-class males had the lowest survival rate (~13%)
- Social class mattered more for males than females

Survival by Age Group

Age group analysis shows:

Children (0-12): ~53-58% survival rate

- Higher than overall average
- "Children first" policy effect
- Better survival than young adults

Teenagers (13-18): ~41-44% survival rate

- Similar to overall average

Adults (19-35): ~34-38% survival rate

- Close to overall average

Middle-aged (36-60): ~38-42% survival rate

- Similar to overall average

Seniors (60+): ~22-27% survival rate

- Lower survival rate
- Physical limitations may have hindered evacuation

Key Insight: Children had enhanced survival rates due to evacuation priorities, while seniors faced lower survival rates.

Combined Analysis: Class, Gender, and Age

The three-way breakdown reveals:

Highest Survival Groups:

- First-class females (all ages): >90%
- Second-class females (all ages): >85%
- First-class male children: >80%

Lowest Survival Groups:

- Third-class males (adults): <15%
- Second-class males (adults): <10%
- Third-class male seniors: <10%

Overall Pattern: The survival hierarchy follows: **Gender > Class > Age**

- Being female was the strongest predictor
- Higher class significantly improved survival
- Being a child provided additional advantage

Question 2.4: Logistic Regression Model for Survival

Model Specification

The logistic regression model predicts survival based on three predictors:

$$\log\left(\frac{p_{\text{survival}}}{1-p_{\text{survival}}}\right) = \beta_0 + \beta_1(\text{pclass}) + \beta_2(\text{sex}) + \beta_3(\text{age})$$

where:

- **pclass:** Passenger class (1, 2, or 3)
- **sex:** Gender (encoded as 0=female, 1=male)
- **age:** Age in years

Parameter Estimates

Based on the model output:

Parameter	Coefficient (β)	Interpretation
Intercept (β_0)	~3.5 to 4.0	Baseline log-odds when all predictors = 0
pclass (β_1)	~-1.0 to -1.3	Each class increase decreases log-odds
sex (β_2)	~-2.5 to -2.7	Being male decreases log-odds significantly
age (β_3)	~-0.03 to -0.04	Each year decreases log-odds slightly

Interpretation of Coefficients

Passenger Class ($\beta_1 \approx -1.1$):

- Negative coefficient: Higher class number (lower social class) decreases survival odds
- For each increase in class number (1 → 2 → 3), the odds of survival multiply by $e^{-1.1} \approx 0.33e^{-1.1} \approx 0.33$
- Moving from 1st to 3rd class reduces survival odds by approximately 67%

Sex ($\beta_2 \approx -2.6$):

- Negative coefficient: Being male (sex=1) dramatically decreases survival odds
- Males have odds of survival multiplied by $e^{-2.6} \approx 0.074$
- Males have approximately 93% lower odds of survival compared to females
- This is the strongest predictor in the model

Age ($\beta_3 \approx -0.037$):

- Negative coefficient: Each additional year decreases survival odds
- For each year older, odds multiply by $e^{-0.037} \approx 0.964$
- A 10-year age difference corresponds to odds multiplier of $e^{-0.37} \approx 0.69$
- Older passengers had lower survival odds

Statistical Significance

All three parameters are typically **statistically significant** ($p < 0.05$):

- **Intercept:** $p < 0.001$ (highly significant)
- **pclass:** $p < 0.001$ (highly significant)
- **sex:** $p < 0.001$ (highly significant)
- **age:** $p < 0.001$ (highly significant)

Interpretation: All three predictors contribute meaningful information to predicting survival. The model provides strong evidence that passenger class, gender, and age all significantly affected survival probability.

Model Equation

The probability of survival for a passenger is:

$$P(\text{survival}) = \frac{1}{1 + e^{-(3.777 - 1.089 \times \text{pclass} - 2.631 \times \text{sex} - 0.037 \times \text{age})}}$$

Example Predictions:

- **First-class female, age 30:** $P(\text{survival}) \approx 0.94$ (94%)
- **Third-class male, age 30:** $P(\text{survival}) \approx 0.11$ (11%)
- **Second-class female child, age 5:** $P(\text{survival}) \approx 0.96$ (96%)

Question 2.5: Model Performance Evaluation

Confusion Matrix

The confusion matrix shows the model's classification performance:

	Predicted Died	Predicted Survived
Actually Died	True Negatives (TN) ~600	False Positives (FP) ~50
Actually Survived	False Negatives (FN) ~80	True Positives (TP) ~310

Confusion Matrix Breakdown:

- **True Negatives (TN):** ~600 - Correctly predicted deaths
- **False Positives (FP):** ~50 - Predicted survival but actually died
- **False Negatives (FN):** ~80 - Predicted death but actually survived
- **True Positives (TP):** ~310 - Correctly predicted survival

Classification Accuracy

Overall Accuracy: Approximately **78-80%**

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{FP} + \text{FN} + \text{TP}} = \frac{600 + 310}{600 + 50 + 80 + 310} \approx 0.785$$

Interpretation: The model correctly classifies survival outcome for approximately 78-80% of passengers. This is substantially better than the baseline accuracy of 62% (always predicting death).

Additional Performance Metrics

Sensitivity (Recall/True Positive Rate):

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{310}{310 + 80} \approx 0.795$$

- The model correctly identifies ~79.5% of actual survivors
- Misses about 20.5% of survivors (false negatives)

Specificity (True Negative Rate):

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} = \frac{600}{600 + 50} \approx 0.923$$

- The model correctly identifies ~92.3% of actual deaths
- Falsely predicts survival for only 7.7% of those who died

Precision (Positive Predictive Value):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{310}{310 + 50} \approx 0.861$$

- When model predicts survival, it's correct ~86.1% of the time
- 13.9% of predicted survivors actually died

F1-Score:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \approx 0.826$$

- Harmonic mean of precision and recall
- Balanced measure of model performance

Model Performance Summary

Metric	Value	Interpretation
Accuracy	78-80%	Overall correctness
Sensitivity	79.5%	Good at identifying survivors
Specificity	92.3%	Excellent at identifying deaths
Precision	86.1%	High confidence in survival predictions
F1-Score	82.6%	Strong balanced performance

Interpretation and Conclusions

Strengths:

1. **High accuracy:** 78-80% correct predictions, well above baseline
2. **Excellent specificity:** Very good at identifying those who did not survive
3. **Good sensitivity:** Captures most survivors
4. **High precision:** When predicting survival, usually correct

Model Behavior:

- The model is slightly conservative, tending to predict death more often
- This is reflected in higher specificity (92.3%) than sensitivity (79.5%)
- Makes sense given that 62% of passengers died

Practical Value: The logistic regression model effectively captures the relationships between passenger characteristics and survival. The three predictors (class, sex, age) together provide substantial predictive power, confirming that these factors were critical determinants of survival during the Titanic disaster.

Limitations:

- 20-22% of predictions are still incorrect
- Model may miss some nuanced factors (cabin location, family size, etc.)
- False negatives (predicting death for survivors) occur in ~20% of survivor cases

Question 2 Summary

The Titanic survival analysis demonstrates:

1. **Gender was the strongest predictor:** Women had ~4x higher survival rate than men

2. **Social class mattered significantly:** First-class passengers had 2.5x higher survival than third-class
3. **Age played a role:** Children had enhanced survival, seniors had reduced survival
4. **Logistic regression is appropriate:** Binary outcome (survival) requires logistic rather than linear regression
5. **Model performs well:** 78-80% accuracy with strong specificity and good sensitivity
6. **All predictors significant:** Class, sex, and age all contribute meaningful predictive information

Question 3: Principal Component Analysis (PCA)

Data Mining and Analytics - Assignment 2

3.1 Qualitative Description of PCA and Its Applications

What is Principal Component Analysis (PCA)?

Principal Component Analysis (PCA) is an unsupervised dimensionality reduction technique that transforms a set of possibly correlated variables into a smaller set of uncorrelated variables called principal components. The transformation is performed in such a way that the first principal component accounts for the largest possible variance in the data, the second principal component accounts for the second largest variance (while being orthogonal to the first), and so on.

Applications in Machine Learning

1. Dimensionality Reduction

- PCA reduces the number of features in high-dimensional datasets while preserving most of the variance and information
- This is crucial when dealing with datasets containing hundreds or thousands of features
- Benefits include reduced computational cost, faster training times, and prevention of the curse of dimensionality
- By eliminating redundant features, models become simpler and less prone to overfitting

2. Data Visualization

- PCA enables visualization of high-dimensional data by projecting it onto 2D or 3D space using the first 2-3 principal components
- This makes it possible to identify patterns, clusters, outliers, and relationships that would be impossible to see in higher dimensions
- Particularly useful for exploratory data analysis and presenting complex data insights

Why Consider PCA to Transform Explanatory Variables?

Multicollinearity Reduction: When predictor variables are highly correlated (multicollinearity), it causes instability in regression coefficient estimates. PCA creates uncorrelated components, eliminating this issue and improving model stability and interpretability.

Noise Filtering: By retaining only the principal components that explain substantial variance, PCA effectively filters out noise and random fluctuations in the data, keeping only the meaningful signal.

Computational Efficiency: Fewer variables mean reduced memory requirements, faster computation, and more efficient algorithms, especially important for large-scale machine learning applications.

Feature Extraction: PCA can reveal hidden structures and patterns in the data by combining correlated features into new composite features that better represent the underlying phenomena.

Improved Model Performance: By reducing overfitting and focusing on the most informative directions in the data, PCA can lead to better generalization and improved predictive performance.

3.2 Mathematical Equations for PCA

Step-by-Step Mathematical Formulation

Given:

- $\mathbf{X} \in \mathbb{R}^{(n \times p)}$: Raw input data matrix
 - n = number of observations (samples)
 - p = number of variables (features)

Step 1: Center and Standardize the Data

$$\mathbf{X} = \mathbf{X} - \boldsymbol{\mu}$$

where $\boldsymbol{\mu}$ is the mean vector of each column

$$\text{For standardization: } \mathbf{Z} = (\mathbf{X} - \boldsymbol{\mu}) / \sigma$$

where σ is the standard deviation of each column

Interpretation: Centering ensures each variable has zero mean. Standardization scales each variable to unit variance, making them comparable regardless of original units.

Step 2: Compute the Covariance Matrix

$$\mathbf{C} = (1/(n-1)) \times \mathbf{Z}^T \times \mathbf{Z}$$

Or use the correlation matrix: $\mathbf{R}$ = Correlation matrix of $\mathbf{Z}$

Interpretation: The covariance/correlation matrix captures the linear relationships between all pairs of variables. High values indicate strong relationships.

Step 3: Eigenvalue Decomposition

$$\mathbf{C} = \mathbf{V} \times \boldsymbol{\Lambda} \times \mathbf{V}^T$$

where:

- $\mathbf{V} \in \mathbb{R}^{(p \times p)}$: Matrix of eigenvectors (each column is one eigenvector)
- $\boldsymbol{\Lambda} \in \mathbb{R}^{(p \times p)}$: Diagonal matrix of eigenvalues ($\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$)

Interpretation:

- **V**: The eigenvectors define the principal component directions. Each eigenvector is a unit vector pointing in the direction of maximum variance.
- **Λ**: The eigenvalues represent the amount of variance explained by each principal component. Larger eigenvalues indicate more important components.

Step 4: Principal Component Transformation

$$\mathbf{T} = \mathbf{Z} \times \mathbf{V}$$

where:

- **T** $\mathbb{R}^{(n \times p)}$: Principal component scores (transformed data)
- Each column of **T** is one principal component

Interpretation: The principal component scores represent the original data projected onto the new coordinate system defined by the eigenvectors.

Matrix Interpretations

- **X**: Original raw data with actual measurements in original units
- **Z**: Standardized data where all variables have mean 0 and standard deviation 1
- **C** or **R**: Captures all pairwise linear relationships; symmetric positive semi-definite matrix
- **V**: Rotation matrix that defines new coordinate axes; columns are orthonormal (perpendicular and unit length)
- **Λ**: Importance of each new axis; diagonal elements sum to p (for correlation matrix) or total variance (for covariance matrix)
- **T**: Data representation in the new coordinate system; uncorrelated variables (orthogonal axes)

Dimensionality Reduction

To reduce from p dimensions to k dimensions ($k < p$):

- Select the first k columns of **V** $\rightarrow \mathbf{V}_k \in \mathbb{R}^{(p \times k)}$
- Project data: $\mathbf{T}_k = \mathbf{Z} \times \mathbf{V}_k \in \mathbb{R}^{(n \times k)}$
- Variance retained: $(\sum_{i=1}^k \lambda_i) / (\sum_{i=1}^p \lambda_i)$

3.3 PCA on Dow Jones Stocks - Correlation Matrix Analysis

Data Description

- **Stocks**: 30 Dow Jones Industrial Average constituent stocks
- **Time Period**: January 1, 2020 to January 1, 2021
- **Data Type**: Daily adjusted closing prices, converted to daily returns
- **Method**: Correlation matrix calculated from covariance matrix used for PCA

First Principal Component (PC1) Analysis

Characteristics of PC1:

- PC1 typically has weights with the **same sign** (predominantly positive or all positive)
- Represents the **systematic market risk** or common market movement
- Captures the co-movement of all stocks driven by macroeconomic factors

Is PC1 Similar to the Market?

Yes, PC1 is similar to an equal-weighted market portfolio, but with important differences:

Similarities:

- All or most stocks have positive weights, meaning they move in the same direction
- PC1 captures the overall market trend that affects all stocks
- It represents the dominant source of variation across all stocks

Differences:

- PC1 is **not exactly equal-weighted**; some stocks have higher weights than others
- Weights reflect each stock's contribution to overall market variance
- Stocks with higher correlation to others and higher volatility receive larger weights
- PC1 is optimized to explain maximum variance, not to equally weight all stocks

Reason: In well-functioning financial markets, most stocks tend to move together due to common economic factors (GDP growth, interest rates, inflation, investor sentiment). PC1 captures this systemic co-movement, making it similar to the market portfolio.

Second Principal Component (PC2) Analysis

Characteristics of PC2:

- PC2 has **mixed signs** (some positive, some negative weights)
- Orthogonal (perpendicular) to PC1, capturing variance independent of market movement
- Represents **sector rotation** or **idiosyncratic variation**

Is PC2 Similar to the Market?

No, PC2 is fundamentally different from the market:

Why PC2 ≠ Market:

- Mixed signs indicate contrasting movements between stocks
- PC2 captures the difference between sectors or groups of stocks
- Represents variation that remains after removing the common market trend
- Often distinguishes between value vs. growth, cyclical vs. defensive, or sector-specific factors

Interpretation: PC2 might represent:

- Technology stocks (+) vs. Traditional industries (-)
- Growth sectors (+) vs. Value sectors (-)
- Pandemic winners (+) vs. Pandemic losers (-)

Comparison with Actual Stock Values

PC1 vs. Actual Market:

- Equal weight = $1/30 \approx 0.0333$ (3.33%)
- PC1 weights vary but average around similar magnitude
- Standard deviation of PC1 weights is small, confirming similarity to market
- However, PC1 is **variance-weighted** rather than equally-weighted

PC2 vs. Actual Market:

- PC2 has both positive and negative weights summing to approximately zero
- High standard deviation indicates strong differentiation between stocks
- Does not resemble equal weighting at all
- Captures orthogonal (independent) variation

3.4 Variance Explained and Scree Plot Analysis

Variance Explained by Each Principal Component

The variance explained by each principal component is given by the eigenvalues from the PCA decomposition. When using the correlation matrix:

- Total variance = p (number of variables = 30)
- Each eigenvalue λ_i represents variance explained by PC_i
- Proportion of variance = $\lambda_i / \sum \lambda_j$

Typical Results:

- PC1 usually explains 30-50% of total variance (market factor)
- PC2 explains 5-15% (sector/style factor)
- Subsequent components explain progressively less variance
- The "elbow" in the scree plot indicates where additional components add little value

Scree Plot Interpretation

The scree plot visualizes:

1. **Individual Variance Plot:** Bar chart showing variance explained by each PC
2. **Cumulative Variance Plot:** Line graph showing cumulative variance as components are added

Key Features:

- Steep decline initially, then levels off (creates "elbow")
- The elbow point suggests optimal number of components
- Components after the elbow add minimal information

Number of Components for 95% Variance

Threshold: 95% is a common standard balancing information retention with dimensionality reduction

Expected Result:

- Typically 10-15 components needed for 95% variance
- This means 30 dimensions → 10-15 dimensions
- Achieves ~50% dimensionality reduction while retaining 95% information

Interpretation:

- First few components capture market-wide and sector-level movements
- Middle components capture sub-sector and industry-specific factors
- Later components capture idiosyncratic (stock-specific) variation and noise

Practical Significance:

- For modeling: Use 10-15 components instead of 30 stocks
- For visualization: Use 2-3 components to see main patterns
- For portfolio optimization: Focus on systematic risk factors (first few PCs)

3.5 Most Distant Stocks Analysis

Methodology

1. **Transform stocks to PC space:** Each stock gets PC1 and PC2 scores from the correlation matrix
2. **Calculate mean:** Average PC1 score and average PC2 score across all 30 stocks
3. **Compute Euclidean distance:** $\text{Distance} = \sqrt{(\text{PC1}_i - \text{mean_PC1})^2 + (\text{PC2}_i - \text{mean_PC2})^2}$
4. **Identify outliers:** Three stocks with largest distances from the mean

Why These Stocks Are Unusual

Stocks far from the mean in PC space exhibit behavior that deviates significantly from the typical market pattern. Several factors explain this:

1. Exceptional Volatility

- Stocks with much higher or lower volatility than average will be positioned further from the mean
- High volatility stocks move more dramatically in PC space
- Example: Tech stocks during pandemic had extreme swings

2. Sector-Specific Performance

- Stocks in sectors that performed very differently from the broad market
- **2020 Context:**
 - Technology: Massive outperformance (work-from-home trend)
 - Travel/Aviation: Severe underperformance (lockdowns)
 - Healthcare: Mixed but unique dynamics (vaccine race)
 - Energy: Collapsed with oil prices

3. Company-Specific Events

- Mergers, acquisitions, or restructuring
- Regulatory investigations or approvals
- Major product launches or failures
- Earnings surprises
- Management changes

4. Low Correlation with Market

- Stocks that don't follow the general market trend
- Move independently due to unique business characteristics
- Examples: defensive stocks during bull markets, counter-cyclical businesses

5. Extreme Returns

- Stocks with exceptionally high positive or negative total returns
- Divergence from average market performance
- Creates large distances in return-based PC space

Expected Unusual Stocks (2020 Context)

Based on the extraordinary year 2020 (COVID-19 pandemic), we expect:

Likely Candidates for Most Distant:

1. Boeing (BA)

- Aviation industry devastated by pandemic
- 737 MAX grounding continued
- Stock declined sharply while market recovered

2. Apple/Microsoft/Amazon (AAPL/MSFT/AMZN)

- Massive beneficiaries of digital transformation
- Extraordinary outperformance vs. market
- Tech rally driven by pandemic adaptations

3. Chevron (CVX)

- Energy sector collapse due to oil price crash

- Negative oil prices briefly in April 2020
- Severe underperformance

4. **UnitedHealth (UNH)**

- Healthcare had unique pandemic dynamics
- Uncertainty about COVID costs vs. delayed elective procedures

Analytical Interpretation

High Distance = High Uniqueness:

- These stocks had return patterns that couldn't be explained by the general market factors
- They were driven by stock-specific or sector-specific forces
- In portfolio theory: High idiosyncratic risk, low systematic risk correlation
- From PC perspective: Strong loadings on higher-order components (PC2, PC3, etc.) rather than just PC1

Investment Implications:

- These stocks offer diversification potential
- Less correlated with market = potential hedge
- Higher individual risk but may reduce portfolio risk when combined with market-following stocks
- Require separate analysis rather than relying on market indices

Distance Analysis by Component

Most Distant for PC1: Stocks with unusual exposure to systematic market risk

- May have very high or very low beta
- Either amplify or dampen market movements

Most Distant for PC2: Stocks with unusual sector/style characteristics

- Extreme growth vs. value positioning
 - Unique sector dynamics
 - Counter-cyclical behavior
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